



National
Qualifications
2024

2024 Physics

Advanced Higher

Question Paper Finalised Marking Instructions

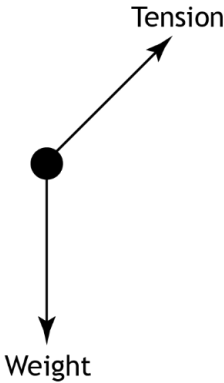
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Marking Instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		$v = 2.4 + 13t - 0.69t^2$ $a = \frac{dv}{dt} = 13 - 1.38t \quad (1)$ $\text{at } t = 2.0$ $a = 13 - (1.38 \times 2.0) \quad (1)$ $a = 10 \text{ ms}^{-2} \quad (1)$	3	Accept: 10.2, 10.24
	(b)		$s = \left(\int v dt = \right) 2.4t + \frac{13}{2}t^2 - \frac{0.69}{3}t^3 (+c) \quad (1)$ $(\text{at } t = 0, s = 0 \therefore c = 0)$ $s = (2.4 \times 2.0) + \left(\frac{13}{2} \times 2.0^2 \right) - \left(\frac{0.69}{3} \times 2.0^3 \right) \quad (1)$ $s = 29 \text{ m} \quad (1)$	3	Accept: 30, 29.0, 28.96 Alternative limits method $\left(s = \int_0^2 (2.4 + 13t - 0.69t^2) dt \right)$ $s = \left[2.4t + \frac{13t^2}{2} - \frac{0.69t^3}{3} \right]_0^2 \quad (1)$ $s = \left(2.4 \times 2.0 + \frac{13 \times 2.0^2}{2} - \frac{0.69 \times 2.0^3}{3} \right) (-0) \quad (1)$ $s = 29 \text{ m} \quad (1)$

Question			Expected response	Max mark	Additional guidance
2.	(a)		$v = r\omega$ $v = 0.419 \times 3.01$ $v = 1.26 \text{ ms}^{-1}$	2	SHOW question If final line is missing max 1.
	(b)	(i)		2	Accept: mg or W for weight Accept: T for tension 1 mark for each correctly labelled force with correct direction. If centripetal force is shown correctly - ignore If centripetal force is shown incorrectly - max 1 mark Any mention of centrifugal force - award 0 marks
		(ii)	$\left(T \sin \theta = \frac{mv^2}{r} \quad \text{and} \quad T \cos \theta = mg \right)$ $\frac{T \sin \theta}{T \cos \theta} = \frac{\left(\frac{mv^2}{r} \right)}{mg}$ $\tan \theta = \frac{v^2}{gr}$	(1,1) 2	Not a standard show question 1 mark for both components of tension 1 mark for dividing Accept: $\left(\tan \theta = \frac{F_c}{W} \right)$ $\tan \theta = \frac{\left(\frac{mv^2}{r} \right)}{mg} \quad (1, 1)$ $\tan \theta = \frac{v^2}{gr}$ 1 mark for both forces F_c and W 1 mark for dividing
		(iii)	$\tan \theta = \frac{v^2}{gr}$ $\tan \theta = \frac{1.26^2}{9.8 \times 0.419}$ $\theta = 21^\circ$	(1) (1) 2	Accept: 20, 21.1, 21.14 Accept answers in radians

Question			Expected response	Max mark	Additional guidance
2.	(c)		θ decreases (1) (As $\tan \theta = \frac{v^2}{gr}$) v and r both decrease but <u>v is squared.</u> (1)	2	MUST JUSTIFY Accept: θ decreases W stays the same and F_c decreases

Question			Expected response	Max mark	Additional guidance
3.	(a)		$I = \left(\frac{1}{3}ml^2\right) \times 3 \quad \text{or} \quad I = \left(\frac{1}{3}ml^2\right) \quad (1)$ $I = \left(\frac{1}{3} \times 2.0 \times 10^4 \times 54^2\right) \times 3 \quad (1)$ $I = 5.8 \times 10^7 \text{ kg m}^2$	2	SHOW question 1 for relationship 1 for all substitutions, including $\times 3$ $\times 3$ can appear at any stage of the calculation $I_{(blade)} = \left(\frac{1}{3}ml^2\right)$ $I_{(blade)} = \left(\frac{1}{3} \times 2.0 \times 10^4 \times 54^2\right)$ $I_{(total)} = \left(\frac{1}{3} \times 2.0 \times 10^4 \times 54^2\right) \times 3$ $I_{(total)} = 5.8 \times 10^7 \text{ kg m}^2$
	(b)	(i)	$\omega = \omega_0 + \alpha t \quad (1)$ $0 = 3.7 + (\alpha \times 550) \quad (1)$ $\alpha = -6.7 \times 10^{-3} \text{ rad s}^{-2} \quad (1)$	3	Accept: -7, -6.73, -6.727
		(ii)	$\tau = I\alpha \quad (1)$ $\tau = 5.8 \times 10^7 \times (-)6.7 \times 10^{-3} \quad (1)$ $\tau = (-)3.9 \times 10^5 \text{ N m} \quad (1)$	3	Accept: 4, 3.89, 3.886 Or consistent with (b)(i)
	(c)	(i)		1	
		(ii)	The mass of the ice is negligible.	1	Alternatives: Change in moment of inertia (of the system) is negligible.
	(d)		$P = 0.3\rho Av^3$ $P = 0.3 \times 1.29 \times \pi \times 54^2 \times 11.5^3 \quad (1)$ $P = 5.4 \times 10^6 \text{ W} \quad (1)$	2	Accept: 5, 5.39, 5.392

Question			Expected response	Max mark	Additional guidance
4.	(a)		$\frac{GMm}{r^2} = mr \left(\frac{2\pi}{T} \right)^2 \quad (1)$ $\frac{6.67 \times 10^{-11} \times M}{(1.2 \times 10^3)^2} = 1.2 \times 10^3 \left(\frac{2\pi}{11.9 \times 60 \times 60} \right)^2 \quad (1)$ $M = 5.6 \times 10^{11} \text{ kg} \quad (1)$	3	Accept: 6, 5.57, 5.573 $\frac{GMm}{r^2} = mr\omega^2 \text{ and } \omega = \frac{2\pi}{T} \quad (1)$
	(b)	(i)	$v = \sqrt{\frac{2GM}{r}} \quad (1)$ $v = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.6 \times 10^{11}}{1.2 \times 10^3}} \quad (1)$ $v = 0.25 \text{ ms}^{-1} \quad (1)$	3	Accept: 0.2, 0.250, 0.2495 Or consistent with (a)
		(ii)	Gravitational potential energy is less (as $E_p = -\frac{GMm}{r}$) (1) r has decreased/Dimorphos is now in a lower orbit (1)	2	Accept gravitational potential energy is more negative Any statement about change of mass zero marks.
	(c)		Collision only results in a small deflection. (Large distance required for a change of trajectory greater than the diameter of the Earth).	1	Accept arguments in terms of more time for the deflection to affect the path of the asteroid.

Question			Expected response	Max mark	Additional guidance
5.	(a)	(i)	$r_{\text{schwarzschild}} = \frac{2GM}{c^2} \quad (1)$ $r_{\text{schwarzschild}} = \frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{(3.00 \times 10^8)^2} \quad (1)$ $r_{\text{schwarzschild}} = 8.9 \times 10^{-3} \text{ m} \quad (1)$	3	Accept: 9, 8.89, 8.893
		(ii)	$\Delta t = t_G \left(\frac{1}{\sqrt{1 - \left(\frac{r_{\text{schwarzschild}}}{r} \right)}} - 1 \right)$ $\Delta t = (24 \times 60 \times 60) \times \left(\frac{1}{\sqrt{1 - \left(\frac{8.9 \times 10^{-3}}{6.4 \times 10^6} \right)}} - 1 \right) \quad (1)$ $\Delta t = 6.0 \times 10^{-5} \text{ (s)} \quad (1)$	2	Accept: 6, 6.01, 6.007 Or consistent with (a)(i)
	(b)		Clocks on the satellites have to be slowed by $39 \mu\text{s}$ (per day) (1)	1	Accept reduced by $39 \mu\text{s}$ (per day) Do not accept $-39 \mu\text{s}$ on its own.
	(c)	(i)	$a_r = r\omega^2 \quad (1)$ $9.8 = \frac{1.8 \times 10^3}{2} \times \omega^2 \quad (1)$ $\omega = 0.10 \text{ rad s}^{-1} \quad (1)$	3	0.1, 0.104, 0.1043 Accept: $m r \omega^2 = m g$ as a starting point
		(ii)	(Clock) at the rim (would run slower). (1) It experiences a greater <u>acceleration</u> OR It experiences the same <u>effect</u> as being in a stronger gravitational field OR <u>Equivalent</u> to being in a stronger gravitational field (1)	2	MUST JUSTIFY Accept stronger <u>artificial</u> gravity for second mark.

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	$\frac{P}{A} = \sigma T^4$ (1) $\frac{P}{A} = 5.67 \times 10^{-8} \times 2800^4$ (1) $\frac{P}{A} = 3.5 \times 10^6 \text{ W m}^{-2}$ (1)	3	Accept: 3, 3.49, 3.485
		(ii)	$L = 4\pi r^2 \sigma T^4$ (1) $4.38 \times 10^{23} = 4\pi \times r^2 \times 5.67 \times 10^{-8} \times 2800^4$ (1) $r = 1.0 \times 10^8 \text{ m}$ (1)	3	Accept: 1, 1.00, 1.000 Or consistent with (a)(i) $\frac{P}{A} = 3.5 \times 10^6 (= 5.67 \times 10^{-8} \times 2800^4)$
	(b)	(i)	(electron) neutrino Positron	1	Accept: anti-electron
		(ii)	Deuterium only exists for 1 second (before fusing again)	1	Accept deuterium only exists for a (very) short time
	(c)		Lifetime is greater than the age of the (observable) Universe	1	

Question			Expected response	Max mark	Additional guidance
7.	(a)		Electron diffraction/interference	1	Must include experimental evidence rather than just naming an experiment
	(b)	(i)	$\lambda = \frac{h}{p} \quad (1)$ $\lambda = \frac{6.63 \times 10^{-34}}{1.673 \times 10^{-27} \times 1.50 \times 10^7} \quad (1)$ $\lambda = 2.64 \times 10^{-14} \text{ m} \quad (1)$	3	Accept: 2.6, 2.642, 2.6420
		(ii)	(de Broglie) wavelength too small (for wave-like effects) OR (de Broglie) wavelength very small (so object behaves like a particle)	1	Do not accept: the de Broglie wavelength is smaller than the size of the proton.
	(c)	(i)	$\left(\Delta\lambda = \frac{h}{mc} (1 - \cos\theta) \right)$ $\Delta\lambda = (72.5 - 71.1) \times 10^{-12} \quad (1)$ $(72.5 - 71.1) \times 10^{-12} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 3.00 \times 10^8} (1 - \cos\theta) \quad (1)$ $\theta = 65.0^\circ \quad (1)$	3	Accept: 65, 64.98, 64.982 (72.5 - 71.1) is an independent mark
		(ii)	Decrease (1) $\cos\theta$ has increased (1)	2	JUSTIFY (θ has decreased, so) $1 - \cos\theta$ has decreased Accept justification by calculation

Question			Expected response	Max mark	Additional guidance
8.	(a)	(i)	(Component of) the <u>velocity</u> parallel to the (magnetic) field is constant/ results in no (unbalanced) force/is unaffected by the (magnetic) field. (1) (Component of) the <u>velocity</u> perpendicular to the (magnetic) field results in circular motion/ central force. (1)	2	Independent marks 'Horizontal component', 'vertical component' not acceptable
		(ii)	$F = qvB \text{ or } F = qvB \sin \theta \quad (1)$ $F = 1.60 \times 10^{-19} \times 2.40 \times 10^6 \times 95.0 \times 10^{-3} \times \sin 55.0 \quad (1)$ $F = \frac{mv^2}{r} \text{ or } F = \frac{m(v \sin \theta)^2}{r} \quad (1)$ $F = \frac{1.673 \times 10^{-27} \times (2.40 \times 10^6 \times \sin 55.0)^2}{r} \quad (1)$ $\left(\frac{1.60 \times 10^{-19} \times 2.40 \times 10^6 \times 95.0 \times 10^{-3} \times \sin 55.0}{1.673 \times 10^{-27} \times (2.40 \times 10^6 \times \sin 55.0)^2} \right) \quad (1)$ $r = 0.216 \text{ m} \quad (1)$	5	Accept: 0.22, 0.2164, 0.21639 $r = \frac{mv}{qB} \quad (2)$ $r = \frac{1.673 \times 10^{-27} \times 2.40 \times 10^6 \times \sin 55.0}{1.60 \times 10^{-19} \times 95.0 \times 10^{-3}} \quad (2)$ $r = 0.216 \text{ m} \quad (1)$
	(b)		The radius will increase or the pitch will increase (1) The magnetic induction has reduced (1)	2	JUSTIFY

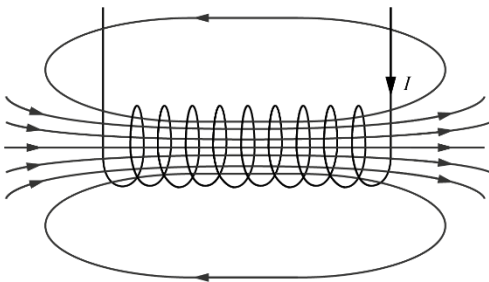
Question			Expected response	Max mark	Additional guidance
9.	(a)		$F = -ky$ (1) $(mg = k\Delta y)$ $510 \times 9.8 = (-)8.60 \times 10^4 \times \Delta y$ (1) $\Delta y = (-)0.058 \text{ m}$ (1)	3	Accept: 0.06, 0.0581, 0.05812
	(b)	(i)	$(F = ma, F = -ky)$ $ma = -ky$ (1, 1) $a = -\omega^2 y$ (1) $((-)m\omega^2 y = (-)ky)$ $\omega = \sqrt{\frac{k}{m}}$	3	Not a standard SHOW question 1 mark for both force relationships 1 mark for equating $a = -\omega^2 y$ on its own 1 mark Final answer not shown max 2 marks
		(ii)	$\omega = \sqrt{\frac{k}{m}}$ $\omega = \sqrt{\frac{8.60 \times 10^4}{510}}$ (1) $\omega = 13 \text{ rad s}^{-1}$ (1)	2	Accept: 10, 13.0, 12.99
		(iii)	Greater (1) The spring constant is greater and the mass remains constant or $\omega \propto \sqrt{k}$ (1)	2	MUST JUSTIFY Accept justification by calculation
	(c)	(i)	Underdamping	1	
		(ii)	Any change that reflects an increase in resistive force: density of oil, area of piston.	1	Do not accept increase the resistive force alone.

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	$v = f\lambda$ (1) $\lambda = \frac{1}{0.18} \text{ (m)}$ (1) $v = 9.5 \times \frac{1}{0.18}$ (1) $v = 53 \text{ ms}^{-1}$ (1)	4	Accept: 50, 52.8, 52.78
		(ii)	$E = kA^2$ (1) $\left(1 = k \times (1.95 \times 10^{-5})^2\right)$ $\frac{1}{8} = \frac{1}{(1.95 \times 10^{-5})^2} \times A^2$ (1) $A = 6.89 \times 10^{-6} \text{ m}$ (1)	3	Accept: 6.9, 6.894, 6.8943 Accept: $\frac{E_1}{A_1^2} = \frac{E_2}{A_2^2}$ (1) $\frac{1}{(1.95 \times 10^{-5})^2} = \frac{0.125}{A_2^2}$ (1) $A_2 = 6.89 \times 10^{-6} \text{ m}$ (1)
		(iii)	The energy/amplitude of the wave becomes too small to be detected.	1	Accept: The energy/amplitude of the wave would reduce to zero.
	(b)		$(d = vt)$ $\Delta t = \frac{0.34}{53} - \frac{0.34}{150}$ (1),(1) $\Delta t = 4.1 \times 10^{-3} \text{ s}$ (1)	3	Or consistent with (a)(i) Accept: 4, 4.15, 4.148 (1) for all substitutions (1) for subtraction (1) for final answer

Question			Expected response	Max mark	Additional guidance
11.	(a)		Helium-neon laser (1) Longest wavelength and $\Delta x = \frac{\lambda D}{d}$, D and d are constant OR Longest wavelength and $\Delta x \propto \lambda$ (1)	2	JUSTIFY
	(b)	(i)	$\Delta x = \frac{0.191}{12}$ (1) $\Delta x = \frac{\lambda D}{d}$ (1) $\frac{0.191}{12} = \frac{\lambda \times 3.15}{1.00 \times 10^{-4}}$ (1) $\lambda = 5.05 \times 10^{-7} \text{ m}$ (1)	4	Accept: 5.1, 5.053, 5.0529 First mark independent
		(ii)	Measurement of separation of the fringes is too small. OR Measurement of slits to screen distance too large. OR Measurement of slit separation too small.	1	
	(c)	(i)	(The multiple tracks on the CD result in multiple) <u>coherent</u> sources of light (multiple reflections) (1) (Bright spots are where) the light meets <u>in phase</u> (to give constructive interference). (1)	2	Accept: division of wavefront in place of coherent Independent marks
		(ii) (A)	Reduce the (percentage/fractional) uncertainty in the measurements (of x and/or h) OR Reduces the (percentage/fractional) uncertainty in the separation of the tracks.	1	Do not accept reduces the scale reading uncertainty. Do not accept reduces the absolute uncertainty in the measurement.
		(ii) (B)	Dimmer spots, wider spots, fewer spots on the screen	1	

Question			Expected response	Max mark	Additional guidance
12.	(a)		Polaroid sheet/analyser (in front of screen) (1) (Observation of) light intensity change when the polaroid sheet / analyser is rotated (1)	2	If mention of two polaroid sheets (both polariser and analyser) - zero marks.
	(b)		Award 3 marks where the candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. Award 2 marks where the candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. Award 1 mark where the candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem. Award 0 marks where the candidate has not demonstrated an understanding of the physics involved. There is no evidence that they have recognised the area of physics involved, or they have not given any statement of a relevant physics principle. Award this mark also if the candidate merely restates the physics given in the question.	3	Candidates may use a variety of physics arguments to answer this question. Award marks based on candidates demonstrating overall good, reasonable, limited, or no understanding.

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	(The electrical) force acting on a unit positive charge.	1	
		(ii)	$E = \frac{Q}{4\pi\epsilon_0 r^2} \quad (1)$ $E = \frac{-4.5 \times 10^{-9}}{4\pi \times 8.85 \times 10^{-12} \times 0.088^2} \quad (1)$ $E = -5.2 \times 10^3 \text{ NC}^{-1} \quad (1)$	3	Accept: -5, -5.23, -5.225 Using 9×10^9 gives -5.230
	(b)	(i)	$(E_T = \sqrt{E_1^2 + E_2^2})$ $E_T = \sqrt{(-5.2 \times 10^3)^2 + (-5.2 \times 10^3)^2} \quad (1)$ $E_T = 7.4 \times 10^3 \text{ NC}^{-1} \quad (1)$	2	Accept: 7, 7.35, 7.354 Or consistent with (a)(ii) Accept alternative methods using trigonometry.
		(ii)	(Vertically) down the page	1	Do not accept down/downwards on its own
	(c)	(i)	$W = QV \quad (1)$ $W = (-)4.5 \times 10^{-9} \times (-)325 \quad (1)$ $(eV) = \frac{W}{1.60 \times 10^{-19}} \quad (1)$ $(eV) = \frac{4.5 \times 10^{-9} \times 325}{1.60 \times 10^{-19}} \quad (1)$ $(eV) = 9.1 \times 10^{12} \text{ (electron volts)} \quad (1)$	4	Accept: 9, 9.14, 9.141
		(ii)	The starting position for Q_2 was not at the zero of electric potential.	1	Accept: Q_2 did not come from infinity.

Question			Expected response	Max mark	Additional guidance
14.	(a)			2	1 mark for shape 1 mark for direction
	(b)		$\left(B = \frac{\mu_0 INR^2}{2(R^2 + x^2)^{\frac{3}{2}}} \right)$ $B = \frac{4\pi \times 10^{-7} \times 3.5 \times 64 \times (120 \times 10^{-3})^2}{2((120 \times 10^{-3})^2 + 0^2)^{\frac{3}{2}}} \quad (1)$ $B = 1.2 \times 10^{-3} \text{ T} \quad (1)$	2	Accept: 1, 1.17, 1.173 Substitution of $x = 0$ not required for first mark
	(c)	(i)	(In a ferromagnetic material) the magnetic <u>dipoles/domains</u> (in the material) can be made to align, (resulting in the material becoming magnetised).	1	
		(ii)	Iron (or nickel or cobalt)	1	Any other ferromagnetic material is acceptable, such as steel or neodymium
		(iii)	The field lines will be more densely/ tightly packed together OR there will be more of them.	1	

Question			Expected response	Max mark	Additional guidance
15.			Award 3 marks where the candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks. Award 2 marks where the candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. Award 1 mark where the candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem. Award 0 marks where the candidate has not demonstrated an understanding of the physics involved. There is no evidence that they have recognised the area of physics involved, or they have not given any statement of a relevant physics principle. Award this mark also if the candidate merely restates the physics given in the question.	3	Candidates may use a variety of physics arguments to answer this question. Award marks based on candidates demonstrating overall good, reasonable, limited, or no understanding.

Question			Expected response	Max mark	Additional guidance
16.	(a)	(i)	Five (time constants)	1	
		(ii)	$\tau = RC$ (1) $\frac{1.0 \times 10^{-3}}{5} = R \times 220 \times 10^{-9}$ (1) $R = 9.1 \times 10^2 \Omega$ (1)	3	Accept: 9, 9.09, 9.091 Or consistent with (a)(i)
	(b)		Equal to. (1) $(\tau = RC)$ and R and C are unchanged (1)	2	MUST JUSTIFY Accept: The time constant for the circuit is independent of the potential difference of the power supply.
	(c)	(i)	The opposition of a capacitor to changing current.	1	Accept: the opposition of a capacitor to AC Do not accept change in direction or change in magnitude on its own
		(ii)	$X_C = \frac{1}{2\pi fC}$ (1) $X_C = \frac{1}{2\pi \times 77 \times 220 \times 10^{-9}}$ (1) $X_C = 9.4 \times 10^3 \Omega$ (1)	3	Accept: 9, 9.40, 9.395

Question			Expected response	Max mark	Additional guidance
17.	(a)	(i)	$\left(\bar{t} = \frac{(16.09+16.26+16.15+16.48+16.22)}{5} \right)$ $\bar{t} = 16.24(\text{s})$ $\left(\bar{T} = \frac{\bar{t}}{10} \right)$ $\bar{T} = 1.624 \text{ s}$	1	Accept: 1.62
		(ii)	$\Delta t = \frac{(16.48 - 16.09)}{5}$ $\left(\Delta t = 0.078 \text{ (s)} \right)$ $\frac{\Delta T}{T} = \frac{\Delta t}{t}$ $\frac{\Delta T}{1.624} = \frac{0.078}{16.24}$ $\Delta T = 0.008 \text{ s}$	2	Suspend sig figs rules Or consistent with (a) (i) Accept: 0.01, 0.0078 $\Delta t = \frac{(1.648 - 1.609)}{5}$ wrong physics zero marks.
	(b)		Acceptable substitutions into gradient relationship.	3	Or consistent with points chosen from the line Evidence of data extracted from the graph must be shown Use of single point only - award 0 marks.
	(c)		$\left(m = \frac{4\pi^2}{g} \right)$ $4.14 = \frac{4\pi^2}{g}$ $g = 9.54 \text{ ms}^{-2}$	2	Or consistent with (b) Independent marks. Accept: points all close to the line of best fit for precision

[END OF MARKING INSTRUCTIONS]

General marking principles for Physics Advanced Higher

Always assign marks for each candidate response in line with these marking principles, the Physics: general marking principles (GMPs) ([Physics: general marking principles - National 3 to Advanced Higher \(sqa.org.uk\)](https://www.sqa.org.uk)) and the detailed marking instructions for this assessment.

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted from a maximum on the basis of errors or omissions.
- (b) If a specific candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Where a candidate incorrectly answers part of a question and carries the incorrect answer forward in the following part, award marks if the incorrect answer has then been used correctly in the subsequent part or 'follow-on'. (GMP 16)
- (d) Award full marks for a correct final answer (including units if required) on its own, unless a numerical question specifically requires evidence of working to be shown, eg in a 'show' question. (GMP 1)
- (e) Award marks where a diagram or sketch correctly conveys the response required by the question. Clear and correct labels (or the use of standard symbols) are usually required for marks to be awarded. (GMP 19)
- (f) Award marks for knowledge of relevant relationships alone. When a candidate writes down several relationships and does not select the correct one to continue with, for example by substituting values, do not award a mark. (GMP 1c)
- (g) Award marks for the use of non-standard symbols where the symbols are defined **and** the relationship is correct, or where the substitution shows that the relationship used is correct. This must be clear and unambiguous. (GMP 20)



- (h) Do not award marks if a 'magic triangle' (eg ) is the only statement in a candidate's response. To gain the mark, the correct relationship must be stated, for example $V = IR$ or $R = \frac{V}{I}$. (GMP 2)
- (i) In rounding to an expected number of significant figures, award the mark for responses that have up to two figures more or one figure less than the number in the data with the fewest significant figures. (GMP 6)

For example:

Data in question is given to 3 significant figures.

Correct final answer is 8.16 J.

Final answer 8.2 J or 8.158 J or 8.1576 J - award the final mark.

Final answer 8 J or 8.15761 J - do not award the final mark

(Note: the use of a recurrence dot, eg $0.\dot{6}$, would imply an infinite number of significant figures and would therefore not be acceptable).

- (j) Award marks where candidates have incorrectly spelled technical terms, provided that responses can be interpreted and understood without any doubt as to the meaning. Where there is ambiguity, do not award the mark. Two specific examples of this would be when the candidate uses a term that might be interpreted as 'reflection', 'refraction' or 'diffraction' (for example 'defraction'), or one that might be interpreted as either 'fission' or 'fusion' (for example 'fussion'). (GMP 22)
- (k) Only award marks for a valid response to the question asked. Where candidates are asked to:
- **identify, name, give, or state**, they need only name or present in brief form.
 - **describe**, they must provide a statement or structure of characteristics and/or features.
 - **explain**, they must relate cause and effect and/or make relationships between things clear.
 - **determine or calculate**, they must determine a number from given facts, figures or information.
 - **estimate**, they must determine an approximate value for something.
 - **justify**, they must give reasons to support their suggestions or conclusions. For example this might be by identifying an appropriate relationship and the effect of changing variables.
 - **show that**, they must use physics [and mathematics] to prove something, for example a given value - *all steps, including the stated answer, must be shown*.
 - **predict**, they must suggest what may happen based on available information.
 - **suggest**, they must apply their knowledge and understanding of physics to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of physics.
 - **use their knowledge of physics or aspect of physics to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates are given credit for the breadth and/or depth of their conceptual understanding.

(I) **Marking in calculations**

Example question

The current in a resistor is 1.5 amperes when the potential difference across it is 7.5 volts. Calculate the resistance of the resistor. (3 marks)

	Example response	Mark and comment
1.	$V = IR$ $7.5 = 1.5R$ $R = 5.0 \Omega$	1 mark: relationship 1 mark: substitution 1 mark: correct answer
2.	5.0Ω	3 marks: correct answer
3.	5.0	2 marks: unit missing
4.	4.0Ω	0 marks: no evidence, wrong answer
5.	$__\Omega$	0 marks: no working or final answer
6.	$R = \frac{V}{I} = \frac{7.5}{1.5} = 4.0 \Omega$	2 marks: arithmetic error
7.	$R = \frac{V}{I} = 4.0 \Omega$	1 mark: relationship only
8.	$R = \frac{V}{I} = __\Omega$	1 mark: relationship only
9.	$R = \frac{V}{I} = \frac{7.5}{1.5} = __\Omega$	2 marks: relationship and substitution, no final answer
10.	$R = \frac{V}{I} = \frac{7.5}{1.5} = 4.0$	2 marks: relationship and substitution, wrong answer
11.	$R = \frac{V}{I} = \frac{1.5}{7.5} = 5.0 \Omega$	1 mark: relationship but wrong substitution
12.	$R = \frac{V}{I} = \frac{75}{1.5} = 5.0 \Omega$	1 mark: relationship but wrong substitution
13.	$R = \frac{I}{V} = \frac{1.5}{7.5} = 5.0 \Omega$	0 marks: wrong relationship
14.	$V = IR$ $7.5 = 1.5 \times R$ $R = 0.2 \Omega$	2 marks: relationship and substitution, arithmetic error
15.	$V = IR$ $R = \frac{I}{V} = \frac{1.5}{7.5} = 0.2 \Omega$	1 mark: relationship correct but wrong rearrangement of symbols

2024 Advanced Higher Physics Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	3	2.8	93.0%	21	0.1%
Q1b	3	2.7	89.5%	30	0.4%
Q2a	2	2	97.6%	19	0.2%
Q2bi	2	0.9	43.5%	34	4.7%
Q2bii	2	1	48.7%	52	10.2%
Q2biii	2	1.8	88.4%	28	0.7%
Q2c	2	0.8	38.8%	29	0.2%
Q3a	2	1.8	87.7%	33	0.2%
Q3bi	3	2.3	77.4%	33	0.4%
Q3bii	3	2.4	80.2%	34	1.0%
Q3ci	1	0.7	65.3%	32	3.2%
Q3cii	1	0.5	50.2%	18	5.5%
Q3d	2	1.5	73.8%	29	0.1%
Q4a	3	1.8	60.0%	52	0.9%
Q4bi	3	2.6	88.3%	36	2.1%
Q4bii	2	0.5	25.8%	25	8.6%
Q4c	1	0.2	23.4%	7	4.3%
Q5ai	3	2.8	92.6%	26	0.7%
Q5aii	2	1.6	78.1%	31	2.6%
Q5b	1	0.4	43.8%	27	6.7%
Q5ci	3	1.5	51.2%	32	4.6%
Q5cii	2	0.7	35.1%	42	3.1%
Q6ai	3	2.4	78.5%	30	1.1%
Q6aii	3	2.7	88.5%	34	2.1%
Q6bi	1	0.6	57.0%	50	2.1%
Q6bii	1	0.3	31.6%	28	10.2%
Q6c	1	0.2	18.8%	21	8.4%
Q7a	1	0.3	29.1%	36	3.0%
Q7bi	3	2.6	86.2%	46	3.1%
Q7bii	1	0.5	54.9%	34	8.1%
Q7ci	3	2	66.0%	49	2.4%
Q7cii	2	1	50.8%	45	7.3%
Q8ai	2	0.7	35.4%	52	6.4%
Q8aii	5	2.9	57.5%	64	5.3%
Q8b	2	0.6	30.7%	34	6.3%
Q9a	3	2.4	80.7%	46	5.7%
Q9bi	3	1.2	41.5%	47	11.9%
Q9bii	2	1.6	80.9%	46	3.8%
Q9biii	2	0.9	44.0%	43	5.1%
Q9ci	1	0.3	32.4%	38	14.0%
Q9cii	1	0.4	40.6%	37	9.4%
Q10ai	4	3	74.9%	59	4.1%
Q10aii	3	1.9	61.7%	64	9.7%
Q10aiii	1	0.6	58.5%	34	8.8%
Q10b	3	1.6	54.0%	51	13.4%
Q11a	2	1.2	61.8%	52	3.0%
Q11bi	4	2.8	70.1%	60	4.3%
Q11bii	1	0.1	10.9%	36	14.0%
Q11ci	2	0.2	9.6%	19	12.8%
Q11ciiA	1	0.3	33.9%	44	9.0%
Q11ciiB	1	0.5	53.0%	32	11.7%
Q12a	2	1.1	57.4%	46	13.3%
Q12b	3	1	31.8%	57	12.3%
Q13ai	1	0.4	43.0%	47	8.5%
Q13aii	3	1.8	61.5%	42	3.3%
Q13bi	2	0.7	35.1%	52	8.3%
Q13bii	1	0.2	17.8%	27	7.1%
Q13ci	4	2.1	51.7%	57	8.6%
Q13cii	1	0.1	10.2%	32	28.3%
Q14a	2	0.2	9.5%	26	12.3%
Q14b	2	1.4	69.9%	41	3.2%
Q14ci	1	0.1	13.7%	33	17.1%
Q14cii	1	0.7	69.6%	37	9.7%
Q14ciii	1	0.4	42.4%	42	14.2%
Q15	3	0.8	27.9%	54	19.2%
Q16ai	1	0.6	56.2%	47	6.4%
Q16aii	3	1.9	63.9%	57	7.5%
Q16b	2	0.7	33.3%	39	9.0%
Q16ci	1	0.3	25.2%	40	19.1%
Q16cii	3	2.2	72.3%	58	10.4%
Q17ai	1	0.2	21.3%	30	2.2%
Q17aii	2	0.7	33.5%	41	11.6%
Q17b	3	1.2	40.8%	46	6.5%
Q17c	2	0.9	42.9%	40	9.9%
Paper	155	88.6	57.2%		

2024 Advanced Higher Physics Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4	5
Q1a	5	1	4	90		
Q1b	7	2	6	85		
Q2a	2	2	97			
Q2bi	42	28	29			
Q2bii	45	12	42			
Q2biii	8	7	85			
Q2c	36	49	14			
Q3a	9	8	84			
Q3bi	15	5	13	67		
Q3bii	12	3	19	66		
Q3ci	35	65				
Q3cii	50	50				
Q3d	21	10	69			
Q4a	24	22	6	49		
Q4bi	3	9	7	81		
Q4bii	67	15	18			
Q4c	77	23				
Q5ai	4	3	5	88		
Q5aii	17	11	73			
Q5b	56	44				
Q5ci	31	24	4	41		
Q5cii	53	23	24			
Q6ai	11	4	24	61		
Q6aii	4	6	11	80		
Q6bi	43	57				
Q6bii	68	32				
Q6c	81	19				
Q7a	71	29				
Q7bi	6	7	9	78		
Q7bii	45	55				
Q7ci	14	24	13	50		
Q7cii	33	32	35			
Q8ai	56	17	27			
Q8aii	11	6	38	6	6	32
Q8b	57	24	19			
Q9a	14	5	4	76		
Q9bi	49	9	10	32		
Q9bii	12	15	73			
Q9biii	42	28	30			
Q9ci	68	32				
Q9cii	59	41				
Q10ai	15	8	2	13	62	
Q10aii	25	16	8	51		
Q10aiii	42	58				
Q10b	39	4	13	44		
Q11a	13	50	37			
Q11bi	15	14	3	12	56	
Q11bii	89	11				
Q11ci	83	15	2			
Q11ciiA	66	34				
Q11ciiB	47	53				
Q12a	35	15	50			
Q12b	37	36	22	5		
Q13ai	57	43				
Q13aii	13	24	28	35		
Q13bi	59	12	29			
Q13bii	82	18				
Q13ci	25	14	20	9	31	
Q13cii	90	10				
Q14a	83	16	2			
Q14b	19	22	59			
Q14ci	86	14				
Q14cii	30	70				
Q14ciii	58	42				
Q15	44	33	18	5		
Q16ai	44	56				
Q16aii	13	31	5	50		
Q16b	63	7	30			
Q16ci	75	25				
Q16cii	17	5	24	55		
Q17ai	79	21				
Q17aii	40	53	7			
Q17b	43	22	5	30		
Q17c	41	32	27			

2024 Advanced Higher Physics Section 3 Project Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.6	55.7%	38
2	4	2.1	51.9%	66
3a	2	1.1	56.2%	48
3b	2	1	50.5%	50
3c	3	2	67.8%	58
4a	1	0.7	65.0%	32
4b	4	2	51.1%	72
4c	3	1.5	50.0%	65
5a	1	0.7	65.8%	39
5b	3	1.5	50.2%	66
5c	3	1	33.5%	55
5d	1	0.5	47.2%	69
6a	1	0.8	83.1%	35
6b	1	0.4	42.4%	43
Paper	30	15.9	53.0%	

Item	Max Mark	0	1	2	3	4
1	1	44	56			
2	4	5	26	34	24	10
3a	2	20	47	33		
3b	2	22	54	23		
3c	3	3	24	40	33	
4a	1	35	65			
4b	4	6	26	35	26	8
4c	3	18	32	32	18	
5a	1	34	66			
5b	3	11	40	37	12	
5c	3	29	46	20	5	
5d	1	53	47			
6a	1	17	83			
6b	1	58	42			
Paper	30					

Item	Max Mark	Item Description
1	1	Brief abstract (summary) stating the overall aim and conclusion
2	3	Underlying chemistry
3a	2	Procedure
3b	1	Safety measures
3c	1	One procedure & second procedure/modification/control/ standardisation
3d	1	Experimental results providing evidence of duplicates
3e	1	Description of the correct use of the appropriate apparatus, chemicals,etc
3f	1	All relevant raw data is recorded
3g	1	Numerical data is appropriately presented
3h	1	Citations and references for three sources of internet/literature data
4a	4	Analysis of data with values calculated correctly using a chemical relationship , scatter, line or bar graph, chromatograms and spectra
4b	1	Calculated values stated to an appropriate number of significant figures
5	1	A valid conclusion that relates to the aim and is supported by data
6	1	A valid comparison of the experimental data with data from the internet/literature sources, or a comparison of duplicate experimental data
7	4	Evaluation including evaluation of data from the internet/literature, evaluative statements supported by justification, quantitative treatment of uncertainties
8	1	A clear and concise report with title, contents page and page numbers